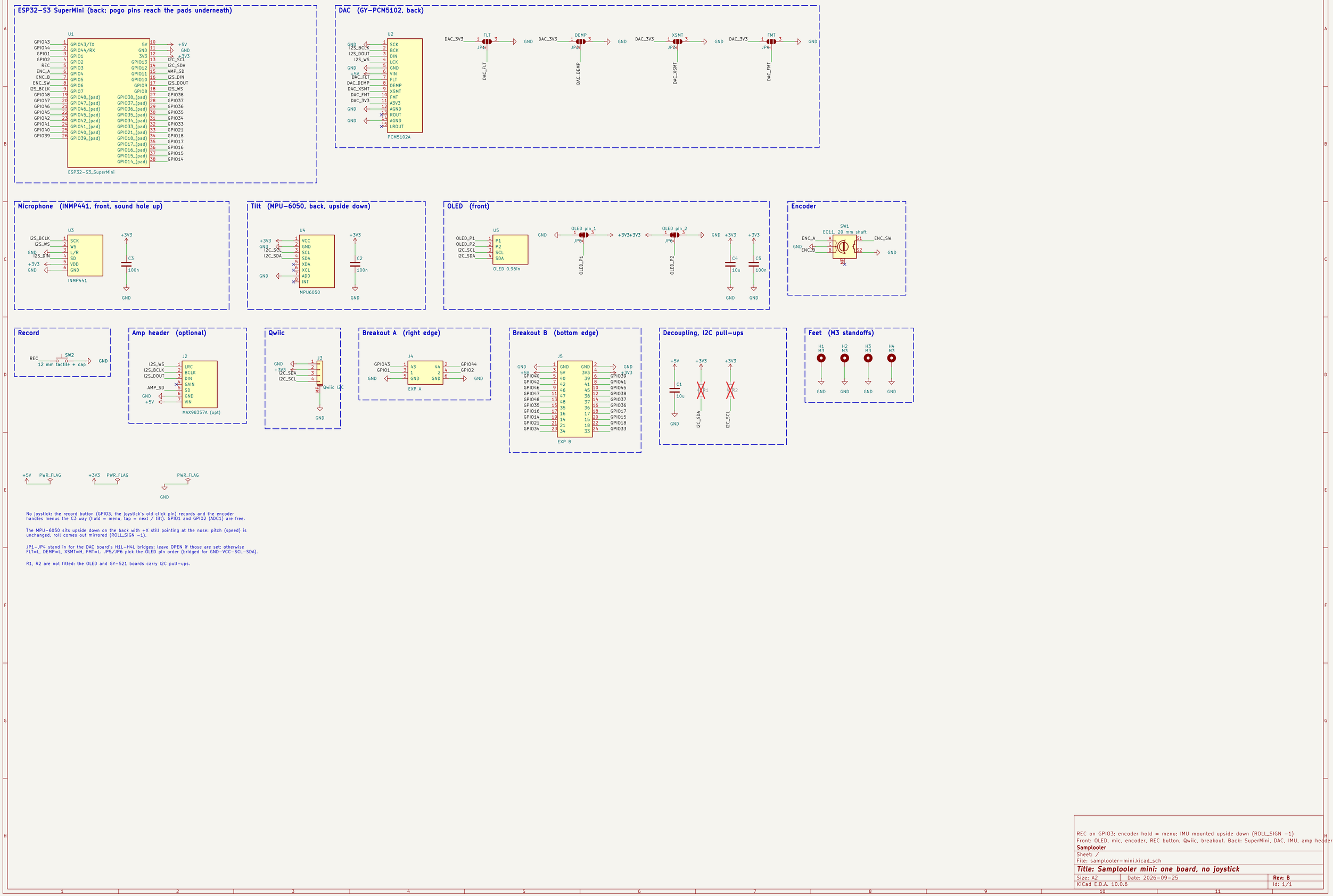

ESP32-S3 SuperMini (back; pogo pins reach the pads underneath)
U1
ESP32-S3_SuperMini
DAC (GY-PCM5102, back)
U2
PCM5102A
FLT
DEMP
XSMT
FMT
JP1
JP2
JP3
JP4
DAC_FLT
DAC_DEMP
DAC_XSMT
DAC_FMT
Microphone (INMP441, front, sound hole up)
U3
INMP441
C3
100n
Tilt (MPU-6050, back, upside down)
U4
MPU6050
C2
100n
OLED (front)
U5
OLED 0.96in
OLED_P1
OLED_P2
JP5
JP6
C4
10u
C5
100n
Encoder
SW1
EC11, 20 mm shaft
Record
SW2
12 mm tactile + cap
Amp header (optional)
J2
MAX98357A (opt)
Qwlic
J3
Qwlic I2C
Breakout A (right edge)
J4
EXP A
Breakout B (bottom edge)
J5
EXP B
Decoupling, I2C pull-ups
C1
10u
R1
R2
Feet (M3 standoffs)
H1
H2
H3
H4
M3
+5V
PWR_FLAG
+3V3
GND
No joystick: the record button (GPIO3, the joystick's old click pin) records and the encoder handles menus the C3 way (hold = menu, tap = next / tilt). GPIO1 and GPIO2 (ADC1) are free.
The MPU-6050 sits upside down on the back with +X still pointing at the nose: pitch (speed) is unchanged, roll comes out mirrored (ROLL_SIGN -1).
JP1-JP4 stand in for the DAC board's H1L-H4L bridges: leave OPEN if those are set; otherwise FLT=L, DEMP=L, XSMT=H, FMT=L. JP5/JP6 pick the OLED pin order (bridged for GND-VCC-SCL-SDA).
R1, R2 are not fitted: the OLED and GY-521 boards carry I2C pull-ups.
REC on GPIO3: encoder hold = menu; IMU mounted upside down (ROLL_SIGN -1)
Front: OLED, mic, encoder, REC button, Qwlic, breakout. Back: SuperMini, DAC, IMU, amp header
Samplooler
Sheet: /
File: samplooler-mini.kicad_sch
Title: Samplooler mini: one board, no joystick
Size: A2
Date: 2026-09-25
Rev: B
KiCad E.D.A. 10.0.6
Id: 1/1